

NIKHIL BAROT

Machine Learning • Generative AI • Finance AI

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PROFESSIONAL SUMMARY

ML Research Engineer with 10+ years of industry experience and active healthcare AI research, combining novel ML architecture design with production deployment expertise. Developed a biologically inspired Neuroidal Network for clinical ECG analysis achieving 99% accuracy on unseen patient data and authored an IEEE-accepted research paper on deep learning framework for safety-critical aerospace predictive maintenance. Built and deployed enterprise-scale LLM applications, RAG pipelines, and AI evaluation frameworks at JPMorgan. Experienced in Python, SQL, PyTorch, statistical modeling, model evaluation, full-cycle AI system development, and cross-functional collaboration with technical and non-technical stakeholders. Passionate about developing evaluation and monitoring methods that ensure safe, equitable, and high-value AI-augmented clinical care.

TECHNICAL SKILLS

Languages: Python, SQL, PL/SQL, Java, TypeScript, C#, Bash/Unix Shell, JavaScript

AI / ML: Neural Networks, Deep Learning, Embedding Models, Clustering (K-Means), Scikit-learn, PyTorch, TensorFlow, Hugging Face Transformers, OpenCV, YOLOv11

AI Evaluation & Observability: Performance Metric Selection, Fairness & Bias Assessment, RAGAS, LangSmith, Phoenix-Arize, Confusion Matrix, Precision/Recall/F1, Confidence Intervals, Bootstrapping

Statistical Modeling: Hypothesis Testing, Power Analysis, Uncertainty Quantification, Sample Weighting, Regression, Time-Series Analysis

GenAI & Agentic AI: LangChain, LangGraph, LlamaIndex, GPT-4, OpenAI API, Google ADK, Prompt Engineering, RAG Pipelines, ReAct Agents

Vector & Enterprise DBs: FAISS, Pinecone, Qdrant, Oracle AI Vector Search, PostgreSQL, MongoDB, DB2, Oracle 11g, Redis

Cloud & MLOps: GCP (BigQuery, Cloud Storage), Oracle Cloud Infrastructure, AWS, Docker, Terraform, MLflow, CI/CD

APIs & Frameworks: FastAPI, REST APIs, Flask, Streamlit, Gradio, WebSockets

Healthcare / Clinical Data: ECG Signal Processing, Waveform Normalization, Clinical Feature Extraction, FHIR (familiarity), Real-World Clinical Data Pipelines

Data Infrastructure: Apache Spark, Kafka, PostgreSQL, Git, Bitbucket, Jira, Agile/Scrum

CERTIFICATIONS

- **Master Google ADK – Agent Development Kit** – Udemy | May 2026
- **Fundamentals of Deep Learning** – NVIDIA | November 2025
- **Oracle Cloud Infrastructure Generative AI Professional** – Oracle | September 2025
- **Oracle Cloud Infrastructure Data Science Professional** – Oracle | September 2025
- **Oracle AI Vector Search Certified Professional** – Oracle | April 2025

WORK EXPERIENCE

Central Michigan University – ML Research • ML Research Engineer

March 2025 – May 2026

- **Invented a Neuroidal Network** for clinical ECG analysis, replaced standard perceptron with biologically inspired neuroidal computational units, encoded arbitrary topologies via adjacency matrices, and built a fully custom non-backpropagation AI architecture with no market equivalent; **provisional patent filed**.
- **Trained the model entirely using Genetic Algorithms (evolutionary optimization)**, achieving **99% accuracy** on unseen ECG patient data with precise R-peak detection; designed and ran end-to-end model evaluation including confusion matrix, precision, recall, and F1 analysis for clinical-grade validation.
- **Built incremental fine-tuning and continuous model improvement pipelines:** pre-trained genetic populations reload and evolve on new patient data without full retraining, directly addressing real-world healthcare deployment requirements (model updating & performance drift management).
- **Validated cross-domain generalizability:** tested the ECG architecture on a human audio binary classification task achieving 76% accuracy, demonstrating transferability of evaluation methodology across clinical data modalities.
- **Preprocessed and cleaned real-world ECG signal data**, handled noise, waveform normalization, and clinical feature extraction; designed data pipelines with rigorous quality controls ensuring clinical-grade model input.
- **Architected Hybrid SAMB-GRU**, a regime-aware dual-head deep learning framework (Conv1D → BiGRU → Temporal Attention) for Remaining Useful Life (RUL) prediction on the NASA C-MAPSS aerospace benchmark; achieved **$R^2 = 0.8825$ and RMSE = 13.48**, outperforming CNN-LSTM, LSTM, GRU, XGBoost, and Random Forest baselines; **accepted at IEEE MLISE (May 2026)**.

- **Designed a regime-aware K-Means clustering preprocessing pipeline** isolating six operational flight regimes with per-cluster normalization; removing it inflated the PHM safety penalty by up to 138%, statistically validating it as the most critical system component using ablation analysis.
- **Built unsupervised diagnostic probes** over a shared latent encoder to attribute subsystem-level fault signatures (Fan/LPC, HPC/Core, Turbine) without fault-label supervision, producing physically coherent degradation patterns across all four NASA C-MAPSS engine subsets.
- **Implemented inverse-RUL sample weighting and leakage-free validation splits** aligned with the asymmetric PHM08 safety objective; reduced FD004 penalty score to $1,041 \pm 96.9$ vs. 2,474 baselines, demonstrating statistical rigor in uncertainty quantification and safety-critical model evaluation.

Office of Information & Technology – Central Michigan University • ML Engineer

May 2025 – Aug 2025

- **Architected and deployed production-grade RAG systems** integrating Pinecone vector databases with LLMs (LangChain, FastAPI), improving document retrieval accuracy by 40% and reducing query response time; implemented evaluation via RAGAS and LangSmith.
- **Built AI safety and monitoring mechanisms** using Guardrails AI and Microsoft Presidio for prompt filtering, PII detection, and system observability, directly supporting reliability, auditability, and compliance of deployed AI systems.
- **Developed UI enhancements and backend updates** for internal service ticket management using C# .NET, TypeScript, and SQL, achieving a 40% reduction in open service tickets; maintained detailed technical documentation throughout.

JPMorgan Chase & Co. – Asset & Wealth Management • Associate – AI/ML Engineer

Aug 2021 – Dec 2024

- **Built a GenAI-powered engineering automation platform** using GPT-4 and LangChain, designed evaluation pipelines with DSPy to benchmark model accuracy, validate response consistency, and measure prompt engineering impact; reduced development cycle time by 60% with 85% generation accuracy rate.
- **Designed LLM evaluation frameworks** to systematically assess AI model performance across enterprise workflows, including accuracy benchmarking, fairness checks, and response quality validation at scale for regulatory compliance.
- **Architected an autonomous multi-agent system** using LangGraph and ReAct agents integrated with Jira, auto-generated test cases, executed them in sandbox environments, and produced structured reports; improved QA engineering productivity by 45%.
- **Implemented AI governance and prompt engineering guardrails** ensuring compliance, auditability, and safety for production LLM deployments at enterprise scale.
- **Led a team of 8 engineers**; mentored junior engineers on AI/ML best practices, presented automation results and live demos to onshore clients across NA and EMEA, and maintained 92% stakeholder satisfaction with 100% on-time delivery.

Oracle (Client: State Street Bank) • Staff Consultant

Oct 2018 – Aug 2021

- **Led end-to-end validation of the ETF-OPT (Exchange-Traded Fund Options Trading) platform** for State Street Corporation, ensuring integrity across order management, trade execution, and settlement workflows; later adopted enterprise-wide by Charles River Development (CRD).
- **Built Python and SQL-based data validation tools** and applied static code analysis using SonarQube across projects, enforcing code quality standards and reducing production defect leakage.

Protean eGov Technologies (Formerly NSDL eGov) • Assistant Manager

Jul 2015 – Sep 2018

- **Managed production deployments for India's National Pension System (NPS) portal**, serving 100,000+ subscribers; authored a Unix shell script reducing deployment time from 2 hours to 20 minutes.
- **Integrated eNPS portal with BillDesk and SBI payment gateways**; tested API integration and validated settlement reconciliation within T+2 days at 99.9% accuracy.

RESEARCH & PUBLICATIONS

- **IEEE MLISE (Accepted May 2026):** "Hybrid SAMB-GRU: A Regime-Aware Deep Learning Framework for Predictive Maintenance", novel architecture outperforming all baselines on NASA C-MAPSS benchmark, with statistical rigor in safety-critical model evaluation.
- **Submitted to (Elsevier)** Neuroidal Network Architecture for Healthcare ECG Analysis, biologically inspired AI system achieving 99% accuracy on clinical cardiac waveform data.

AI & CLOUD PROJECTS

- **Healthcare RAG Prototype (Mutual Fund Advisor):** Built end-to-end RAG pipeline using Python, FastAPI, FAISS vector DB, OpenAI (text-embedding-3-small, GPT-4o-mini), and Streamlit, evaluated retrieval accuracy and response quality end-to-end.
- **Cloud-Native Video Analytics Pipeline (GCP):** Designed and deployed scalable video processing pipeline using OpenCV, Cloud Storage, and BigQuery for large-scale dataset analysis with automated background subtraction.
- **Agentic AI Website Generator:** Built autonomous agent workflows using LangGraph and LLM orchestration capable of independently designing and structuring web pages.

EDUCATION

Central Michigan University

Master of Science in Computer Science

Mount Pleasant, MI

May 2026

Mumbai University

Bachelor of Engineering in Information Technology

Mumbai, India

June 2015

AWARDS & RECOGNITION

- **"The Path Finder" Award** ,JPMorgan Chase, August 2024 (Outstanding Performance & Commitment to Excellence)
- **"The Optimizer" Award** ,JPMorgan Chase, November 2022 & July 2023 (Outstanding Performance & Excellence)
- **"Team Player" Award** ,Oracle (Outstanding Team Contribution)